Non-Ergodic Mess3 Mixtures: How Transformers Represent Hierarchical Belief Structure

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Abstract

We train a small transformer on a non-ergodic dataset of Mess3 processes with different parameters, where each sequence comes from one ergodic component. The model must solve a hierarchical inference problem: identifying the component (meta-synchronization) and tracking belief states within it. We derive geometric predictions connecting this to the Factored World Hypothesis, verify Sierpinski gasket recovery ($R^2 = 0.989$), and analyze synchronization dynamics across positions and layers.

1 Introduction

Language models face non-ergodic data: different documents constitute distinct statistical regimes. In-context learning requires identifying the active regime before predicting optimally. We study this in a controlled setting using Mess3 processes (Marzen and Crutchfield, 2017; Shai et al., 2024), 3-state nonunifilar HMMs whose belief geometry forms a Sierpinski gasket.

By constructing a non-ergodic training set where each sequence comes from one of K Mess3 parameterizations, we create a minimal in-context learning problem: the transformer must determine which process generated the sequence and track its hidden state. This connects to the Factored World Hypothesis (FWH) (Shai et al., 2026): if the model treats component identity and within-component belief as separate factors, it should represent them in orthogonal subspaces with dimensionality $(K-1) + 2 = K+1$, rather than the joint $3K-1$. Crucially, our non-ergodic mixture has *block-diagonal* transition operators (not tensor-product), so the standard FWH does not directly apply. This creates an interesting intermediate case: does the factoring inductive bias extend beyond conditionally-independent processes? The FWH predicts that transformers first learn factored representations, expanding dimensionality only under sustained gradient pressure (Shai et al., 2026). Our setup tests whether this pattern holds for block-diagonal structure.

Contributions. (1) We formalize the non-ergodic Mess3 mixture as hierarchical belief inference and derive geometric predictions, identifying three possible geometries (§4). (2) We verify single-process gasket recovery ($R^2 = 0.989$) and compare model loss against the Bayes-optimal predictor at each position (§5). (3) We analyze synchronization dynamics: how quickly the model identifies the active component across context positions and layers (§6).

2 Background

Mess3 process. A 3-state nonunifilar HMM emitting $z \in \{0, 1, 2\}$, parameterized by emission clarity α and state persistence x, with $\beta = (1-\alpha)/2$, $y = 1-2x$. The labeled transition matrices $T^{(z)}_{ij} = P(\text{emit } z, \text{go to } j \mid \text{state } i)$ are given in Appendix A. The process is nonunifilar: observing a token does not uniquely determine the state transition, so optimal prediction requires maintaining a belief distribution over hidden states. These belief states form a Sierpinski gasket in the 2-simplex (Riechers and Crutchfield, 2018). The spectral gap $\zeta = 1-3x$ controls the exponential decay rate of belief corrections with context distance. The entropy rate $H = -\sum_s \pi(s) \sum_z P(z|s) \log P(z|s)$ with uniform $\pi(s) = 1/3$ gives the theoretical minimum cross-entropy.

Belief geometry in transformers. Transformers linearly represent belief states $\boldsymbol{\eta}^{(z_{1:t})} = \boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)}/\text{norm}$ in their residual stream (Shai et al., 2024). This representation is a constrained Bayesian approximation: $r_1(z_{1:d}) = \boldsymbol{\pi} + \sum_{s=1}^{d} (\boldsymbol{\pi} T^{z_s} T^{d-s} - \boldsymbol{\pi})$, where each correction decays as ζ^{d-s}, reflecting the parallel nature of attention (Piotrowski et al., 2025).

Factored World Hypothesis. Beyond single processes, transformers have an inductive bias toward factored representations (Shai et al., 2026): conditionally independent factors are represented in orthogonal subspaces, requiring $\sum_n (d_n - 1)$ dimensions vs. $\prod_n d_n - 1$ (joint). In synthetic experiments with 5 factors (3 Mess3 + 2 Bloch Walk, joint vocab 432), activations converge to 10 dimensions (factored) rather than 242 (joint). This factoring persists even when it sacrifices fidelity, acting as a representational attractor that models depart from only under sustained gradient pressure.

ICL as source discrimination. Non-ergodicity forces hierarchical inference: identify the source, then predict within it. This is structurally identical to in-context learning in LLMs (Shai et al., 2026). ICL and activation steering both modify the model's belief state about latent concepts, operating additively in log-belief space (Bigelow et al., 2025).

3 Non-Ergodic Mess3 Mixture

3.1 Construction

We construct K Mess3 processes with distinct parameters $\{(\alpha_k, x_k)\}_{k=1}^{K}$. Each training sequence is generated entirely by one component chosen uniformly; the model observes only tokens, not the component identity.

We choose components with well-separated entropy rates H_k and spectral gaps $\zeta_k = 1 - 3x_k$:

	α	x	H_k (nats)	ζ_k
C0	0.95	0.03	0.41	0.91
C1	0.85	0.05	0.69	0.85
C2	0.65	0.15	1.03	0.55

Table 1: Mess3 components. H_k: entropy rate (minimum CE). ζ_k: spectral gap (belief convergence rate). C0 is highly predictable with slow mixing; C2 is nearly uniform with fast mixing.

3.2 Relevance to Language Models

This structure mirrors natural language in three ways. **ICL as source discrimination**: the model must use early tokens to identify the generating process, analogous to topic/style adaptation. **Covariate shift**: different components have different token distributions, requiring in-context adaptation. **Hierarchical latent structure**: component identity is a slow variable (constant per sequence), hidden state is fast (evolves per token), mirroring the multi-scale structure of language.

3.3 Experimental Setup

Model. GPT-2 style decoder-only transformer via TransformerLens (Nanda and Bloom, 2022). Phase 1 (single Mess3): 3 layers, $d_{\text{model}} = 64$, 4 heads. Phase 2 (mixture): 4 layers, $d_{\text{model}} = 128$, 4 heads. Pre-norm, learned positional embeddings, vocabulary {BOS, 0, 1, 2}.

Training. Next-token CE loss, Adam (lr $= 5 \times 10^{-4}$, no weight decay). Fresh batches of 8192 sequences generated on-the-fly. Context: 16 tokens + BOS. Phase 1: 10K steps, Phase 2: 20K steps.

Baselines. Bayes-optimal CE at each position via Monte Carlo: both *oracle* (knows component) and *Bayesian* (infers component from observations).

Analysis. (1) CEV via PCA, (2) linear regression to belief states (R^2), (3) meta-belief regression, (4) per-position/layer probing for synchronization dynamics, (5) attention patterns.

4 Pre-Registered Predictions

4.1 Mathematical Derivation

The joint state space $(k, s) \in \{1, \ldots, K\} \times \{1, 2, 3\}$ has $3K$ states. Since component identity k is fixed within each sequence, the predictive vector factorizes:

$$P(k, s \mid z_{1:t}) = P(k \mid z_{1:t}) \cdot P(s \mid k, z_{1:t}) \quad (1)$$

The meta-belief $P(k \mid z_{1:t})$ is the posterior over components; $P(s \mid k, z_{1:t})$ is the within-component belief. The joint transition operator is block-diagonal: $T^{(z)} = \bigoplus_k P(k) T_k^{(z)}$. However, k is static: after sufficient context $P(k \mid z_{1:t})$ concentrates on the true component.

The meta-belief evolves as $P(k \mid z_{1:t}) \propto P(k) \prod_{\tau=1}^{t} P(z_\tau \mid z_{1:\tau-1}, k)$, where concentration rate depends on KL divergence between component predictive distributions.

4.2 Geometric Predictions

Prediction 1: Dimensionality. Under a factored representation: $(K-1)$ dims for the meta-belief simplex + 2 dims for within-component belief = $K+1$ total. For K=3: factored = 4 dims, joint = 8 dims, separated gaskets $\approx$ 6 dims.

Prediction 2: Context-position dependence. At t=1, meta-belief is near-uniform: geometry is a superposition of K gaskets. As t increases, meta-belief sharpens, geometry resolves into component-specific clusters. Per-position loss should decrease toward the Bayes-optimal rate.

Prediction 3: Layer dependence. Embedding: 3 token vertices. Early layers: component discrimination emerges (meta-synchronization). Later layers: within-component gasket structure sharpens. This ordering reflects computational dependency: identify the process before applying correct transition matrices.

4.3 Multiple Possible Geometries

1. **Separated gaskets**: K distinct gaskets with large offsets encoding component identity ($\sim 2K$ dims). Natural for block-diagonal structure.

2. **Factored (shared belief subspace)**: component identity in $(K-1)$D, shared 2D belief subspace with superimposed gaskets ($K+1$ dims). Strongest evidence for FWH extending beyond tensor-product structure.

3. **Conditional (component-specific belief subspaces)**: each component gets its own oriented 2D subspace ($2K+K-1$ dims).

We predict geometry 1 or 2. Following the inductive bias observed in (Shai et al., 2026), the model may initially learn geometry 1, transitioning toward 2 over training as the factoring attractor pulls representations toward dimensional efficiency.

5 Results

5.1 Phase 1: Single Mess3 Verification

We first verify that our setup reproduces known results: a transformer trained on a single Mess3 process ($\alpha = 0.85$, $x = 0.05$) represents belief states in its residual stream.

The model (3 layers, $d_{\text{model}} = 64$) was trained for 10K steps. Training loss converged to ~ 0.82 nats, matching the Bayes-optimal CE of 0.82 nats (computed via Monte Carlo). The entropy rate is $H = 0.69$ nats; the gap reflects the cost of imperfect belief tracking from finite context (16 tokens).

Linear regression (held-out, 80/20 split) from the final-layer residual stream to ground-truth belief states achieves $R^2 = 0.989$. Layer-wise R^2 improves monotonically: $L0 = 0.984$, $L1 = 0.987$, $L2 = 0.989$, consistent with progressive belief refinement. The recovered geometry (Figure 1) clearly shows the Sierpinski gasket structure.

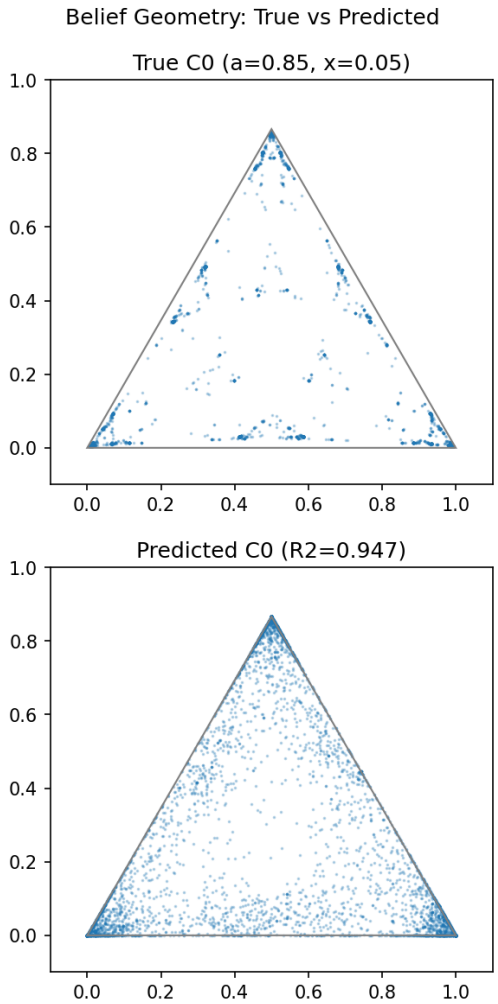

Figure 1: Single Mess3: ground-truth belief geometry (top) and linear regression from residual stream (bottom, $R^2 = 0.947$ at last position). Fractal Sierpinski gasket structure is recovered.

Over training, effective dimensionality (95% CEV) decreases monotonically from 17 to 6, approaching but not reaching the theoretical 2D (the 2-simplex). Additional dimensions encode positional information.

5.2 Phase 2: Non-Ergodic Mixture

We train a larger model (4 layers, $d_{\text{model}} = 128$) on the 3-component non-ergodic mixture (Table 1) for 5K steps.

5.2.1 Belief Regression

All three component-specific Sierpinski gaskets are recovered from the residual stream with high

fidelity (Table 2, Figure 2). The meta-belief (posterior over components) is also linearly decodable with $R^2 = 0.941$.

	R^2 (held-out)	RMSE
C0 (α=0.95, x=0.03)	0.996	0.029
C1 (α=0.85, x=0.05)	0.991	0.039
C2 (α=0.65, x=0.15)	0.989	0.026
Meta-belief	0.941	0.046

Table 2: Belief regression results (held-out 20% test set). Within-component beliefs are nearly perfectly decoded; meta-belief (component identity) is also well-represented.

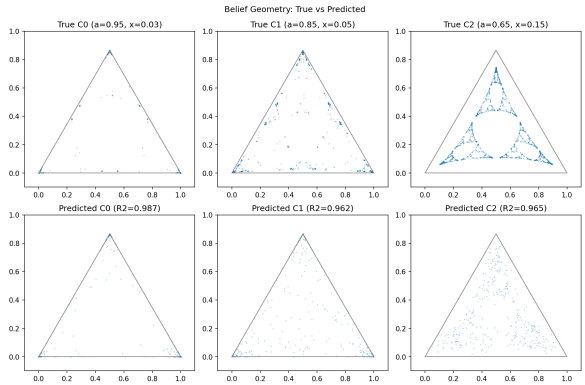

Figure 2: Non-ergodic mixture: ground-truth gaskets (top) vs. predicted from residual stream (bottom) for each component. Each component produces a distinct Sierpinski gasket with different fractal detail, all recovered by linear regression.

5.2.2 Dimensionality

The effective dimensionality (95% CEV) converges to 6 at the final layer, compared to the factored prediction of $K+1 = 4$ and the joint prediction of $3K-1 = 8$. This falls between factored and joint, suggesting partial factoring with some additional dimensions for positional encoding. CEV by layer shows progressive concentration from embedding (2–3 dims) through transformer layers.

5.2.3 Per-Position Loss vs Bayes-Optimal

Figure 3 shows the model's per-position loss against the Bayes-optimal baselines. At position 1, all methods are near-uniform ($\ln 3 = 1.10$ nats). The model's loss tracks the Bayesian baseline (which must infer the component) closely, remaining above the oracle baseline (which knows the component). The gap between model and oracle quantifies the cost of imperfect meta-synchronization.

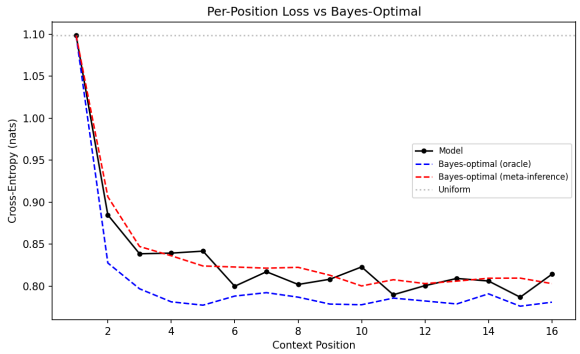

Figure 3: Per-position cross-entropy. The model (black) closely tracks the Bayesian baseline (red, must infer component), remaining above the oracle (blue, knows component). The model achieves near-optimal meta-inference by position ~ 6.

6 Additional Analysis: Synchronization Dynamics

We chose to analyze synchronization dynamics because it directly probes the hierarchical structure of in-context learning: does the model first identify the source, then refine beliefs within it? This question is central to understanding how transformers handle non-ergodic data.

6.1 Layer-Wise Probing

Figure 4 shows R^2 for meta-belief and within-component belief probes at each layer (held-out evaluation).

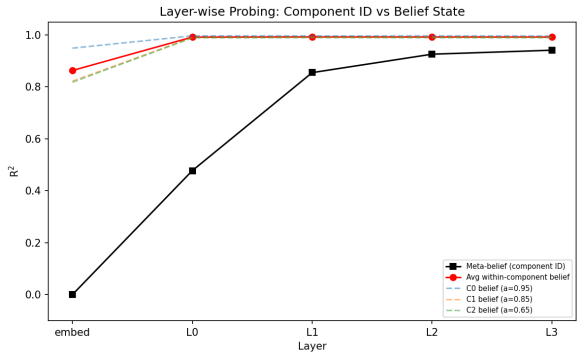

Figure 4: Layer-wise probing reveals that within-component belief (red) is nearly perfect from L0, while meta-belief/component identity (black) builds progressively across layers: $0.00 \to 0.48 \to 0.86 \to 0.93 \to 0.94$.

The key finding: **within-component belief is available before meta-synchronization completes.** The embedding already encodes belief information ($R^2 = 0.86$), and L0 refines it to $R^2 = 0.99$. In contrast, meta-belief (component

identity) starts at chance and builds progressively across all four layers.

This ordering is surprising relative to our Prediction 3, which expected component identification to precede belief refinement. Instead, the model appears to maintain parallel belief tracks for all components simultaneously, with the correct track becoming dominant as meta-synchronization sharpens. This is consistent with the constrained belief update framework: attention computes corrections from each position independently, and these corrections are valid for within-component belief regardless of which component is active.

6.2 Position-Wise Meta-Synchronization

The meta-belief R^2 improves from ~ 0 at position 1 to 0.94 at position 16, with the steepest improvement in positions 2–6. This matches the per-position loss curve (Figure 3): the model achieves near-Bayesian-optimal prediction by position 6, indicating that ~ 5 tokens suffice for reliable component identification with our chosen parameters.

6.3 Attention Patterns

Early-layer attention heads show broad, roughly uniform attention patterns across positions (consistent with evidence gathering for meta-synchronization), while later layers show more structured, position-dependent patterns (consistent with sequential belief updating). Full attention heatmaps are in Appendix.

6.4 What We Would Do With More Time

Given additional time, we would analyze: (1) **Causal interventions**: perturb the component-identity direction in the residual stream and measure the effect on within-component belief geometry. If the representations are truly factored, perturbing one should leave the other intact. (2) **Scaling with K**: how does effective dimensionality scale with the number of components? Our prediction of $K+1$ could be tested with $K = 2, 4, 6$. (3) **Training dynamics**: track when separated gaskets vs. factored representations emerge during training, testing whether the FWH inductive bias applies to block-diagonal structure.

7 Discussion

Non-ergodicity forces hierarchical inference. The model cannot achieve low loss without first identifying the source. This mirrors how LLMs must identify the active regime before predicting well within it.

Block-diagonal structure. The joint operator is block-diagonal, not tensor-product, placing our setup outside the standard FWH setting. Whether the factoring inductive bias extends to this case is an open question that our experiments address.

Residual stream as belief space. The residual stream has no privileged basis; geometric arrangement encodes prediction similarity. If the model discovers that source identity and within-source state are independent dimensions of prediction, it should allocate orthogonal subspaces accordingly.

Limitations. Small model (3–4 layers); static components (no drift or switching); same token alphabet across components. Natural extensions include slowly drifting parameters, GHMM components such as Bloch Walk (which would require belief-state-based generation rather than hidden state simulation), and connections to induction heads (Olsson et al., 2022) as a mechanism for source discrimination.

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A Mess3 Transition Matrices

With $\beta = (1-\alpha)/2$ and $y = 1 - 2x$:

$$T^{(0)} = \begin{pmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{pmatrix} \tag{2}$$

$$T^{(1)} = \begin{pmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{pmatrix} \tag{3}$$

$$T^{(2)} = \begin{pmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{pmatrix} \tag{4}$$

The net transition matrix is:

$$T = T^{(0)} + T^{(1)} + T^{(2)} = \begin{pmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{pmatrix} \tag{5}$$

with uniform stationary distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$ and non-trivial eigenvalue $\zeta = 1 - 3x$.

The emission probability $P(z \mid s) = \sum_j T^{(z)}_{sj}$ gives the entropy rate:

$$H = -\sum_{s=1}^{3} \frac{1}{3} \sum_{z=0}^{2} P(z|s) \log P(z|s) \tag{6}$$

B Bayes-Optimal Loss Derivation

The Bayes-optimal loss at position t is the cross-entropy achieved by a predictor maintaining exact belief states:

$$\mathrm{CE}(t) = \mathbb{E}_{z_{1:t}} \left[-\log P(z_t \mid z_{1:t-1}) \right] \tag{7}$$

where $P(z_t \mid z_{1:t-1}) = \boldsymbol{\eta}^{(z_{1:t-1})} T^{(z_t)} \mathbf{1}$.

For the non-ergodic mixture, we compute two baselines:

- **Oracle**: knows the component, uses component-specific transition matrices.
- **Bayesian**: must infer the component, uses the mixture prediction $P(z_t \mid z_{1:t-1}) = \sum_k P(k \mid z_{1:t-1}) \cdot P(z_t \mid z_{1:t-1}, k)$.

The gap between oracle and Bayesian baselines quantifies the cost of not knowing the component identity.

C Code

All code is available at: `[GitHub repo URL]`. The implementation uses PyTorch with TransformerLens for activation extraction. Key files:

- `src/mess3.py`: Mess3 process generation and belief computation
- `src/bayes_optimal.py`: Bayes-optimal loss computation
- `src/sync_analysis.py`: Synchronization dynamics analysis
- `run.py`: Main experiment script